Lakshya JEE 2.0 2025

Maths

Determinants

DPP: 2

Q1 If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, then value of $\begin{vmatrix} 2a_1+3b_1+4c_1 & b_1 & c_1 \\ 2a_2+3b_2+4c_2 & b_2 & c_2 \\ 2a_3+3b_3+4c_3 & b_3 & c_3 \end{vmatrix}$ is equal to

(A) Δ^2

(B) 4Δ

(C) Δ

(D) 2Δ

Q2 The determinant $D = \begin{vmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) & \cos 2\beta \\ \sin\alpha & \cos\alpha & \sin\beta \\ -\cos\alpha & \sin\alpha & \cos\beta \end{vmatrix}$ independent of

(A) α

(B) β

(C) α and β

(D) Neither α or β

Q3 If $\begin{vmatrix} x^k & x^{k+2} & x^{k+3} \\ y^k & y^{k+2} & y^{k+3} \\ z^k & z^{k+2} & z^{k+3} \end{vmatrix} = \left(x-y\right)\left(y-z\right)\left(z-x\right)\left\{\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right\}$, then:

(A) $k=-2$

(B) $k=-1$

(C) $k=0$

(D) $k=1$

Q4 The number of distinct real roots of $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in the interval $-\frac{\pi}{4} \le x \le \frac{\pi}{4}$ is

(A) 0

(B) 2

(C) 1

(D) 3

Q5 If α, β are non-real numbers satisfying $x^3-1=0$ then the value of $\begin{vmatrix} \lambda+1 & \alpha & \beta \\ \alpha & \lambda+\beta & 1 \\ \beta & 1 & \lambda+\alpha \end{vmatrix}$ is equal to

(A) 0

(B) λ^3

(C) λ^3+1

(D) 4

Q6 If $\begin{vmatrix} x+y & y+z & z+x \\ y+z & z+x & x+y \\ z+x & x+y & y+z \end{vmatrix} = k \begin{vmatrix} x & z & y \\ y & x & z \\ z & y & x \end{vmatrix}$, then k is equal to

(A) -2

(B) 2

(C) -3

(D) 3

Q7 If $\Delta = \begin{vmatrix} 1 & 3\cos\theta & 1 \\ \sin\theta & 1 & 3\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix}$, then the $\frac{1}{1000}\left[\,|\text{maximum value of } \Delta - \text{minimum value of } \Delta|^3\right]$ is equal to _____.

(A) 1

(B) -1

(C) 0

(D) 2

Q8

Given that $xyz=-1$, the value of the determinant $\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix}$,

(A) 0 (B) positive
(C) negative (D) none of these

Q9 If $\Delta=\begin{vmatrix} 0 & b-a & c-a \\ a-b & 0 & c-b \\ a-c & b-c & 0 \end{vmatrix}$, then Δ equals

(A) $a+b+c$ (B) $-(a+b+c)$
(C) abc (D) 0

Q10 If α, β and $\gamma$are the roots of the equation x^3+px+q (with $p\neq 0$ and $q\neq 0$), the value of the determinant $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$, is

(A) p (B) q
(C) p^2-2q (D) None of these

Q11 If

$$f(x)=\begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$$

then $f(500)$ is equal to

(A) 0 (B) 1
(C) 500 (D) -500

Q12 If $\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix}=x+iy$, then

(A) $x=3, y=1$ (B) $x=1, y=3$
(C) $x=0, y=3$ (D) $x=0, y=0$

Q13 If ω is an imaginary cube root of unity, then the value of $\begin{vmatrix} 1+\omega & \omega^2 & -\omega \\ 1+\omega^2 & \omega & -\omega^2 \\ \omega^2+\omega & \omega & -\omega^2 \end{vmatrix}$ is equal to

(A) 0
(B) 2ω
(C) $2\omega^2$
(D) $-3\omega^2$

Q14 If $a_1, a_2, \ldots\ldots, a_{n.}, \ldots\ldots$ are in GP and ai >0 for each i then the determinant

$$\Delta=\begin{vmatrix} \log a_n & \log a_{n+2} & \log a_{n+4} \\ \log a_{n+6} & \log a_{n+8} & \log a_{n+10} \\ \log a_{n+12} & \log a_{n+14} & \log a_{n+16} \end{vmatrix}$$

is equal to

(A) 0 (B) 1
(C) 2 (D) n

Q15 Let $\omega=-\frac{1}{2}+i\frac{\sqrt{3}}{2}$, then the value of the determinant $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1-\omega^2 & \omega^2 \\ 1 & \omega^2 & w^4 \end{vmatrix}$, is

(A) 3ω
(B) $3\omega(\omega-1)$
(C) $3\omega^2$
(D) $3\omega(1-\omega)$

Q16 If a, b, c are in AP, then the value of $\begin{vmatrix} x+2 & x+3 & x+a \\ x+4 & x+5 & x+b \\ x+6 & x+7 & x+c \end{vmatrix}$ is

(A) $x-(a+b+c)$
(B) $9x^2+a+b+c$
(C) $a+b+c$
(D) 0

Q17 If x, y, z are integers in A.P. lying between 1 and 9 and $\mathrm{x51}, \mathrm{y41}$ and $\mathrm{z31}$ are three digit numbers then the value of $\begin{vmatrix} 5 & 4 & 3 \\ x51 & y41 & z31 \\ x & y & z \end{vmatrix}$ is

(A) $x+y+z$
(B) $x-y+z$
(C) 0
(D) None of these

Q18 $\begin{vmatrix} \frac{a^2+b^2}{c} & c & c \\ a & \frac{b^2+c^2}{a} & a \\ b & b & \frac{c^2+a^2}{b} \end{vmatrix}$

is equal to
(A) abc
(B) $2abc$
(C) $4abc$
(D) 0

Q19 $\begin{vmatrix} 2^3 & 3^3 & 3.2^2+3.2+1 \\ 3^3 & 4^3 & 3.3^2+3.3+1 \\ 4^3 & 5^3 & 3.4^2+3.4+1 \end{vmatrix}$ is equal to

(A) 0
(B) 1
(C) 92
(D) None of these

Q20 If $\Delta_1 = \begin{vmatrix} x-y-z & 2x & 2x \\ 2y & y-z-x & 2y \\ 2z & 2z & z-x-y \end{vmatrix}$

and

$\Delta_2 = \begin{vmatrix} x+y+2z & x & y \\ z & y+z+2x & y \\ z & x & z+x+2y \end{vmatrix}$

then

(A) $\Delta_1 = 2\Delta_2$
(B) $\Delta_2 = 2\Delta_1$
(C) $\Delta_1 = \Delta_2$
(D) None of these

Q21 If ω, ω^2 are imaginary cube roots of unity and $\Delta_1 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} 1 & 1 & \omega \\ 1 & 1 & \omega^2 \\ \omega^2 & \omega & 1 \end{vmatrix}$,

then $\frac{\Delta_1}{\Delta_2}$ is equal to
(A) $\sqrt{3}$
(B) $\sqrt{3}i$
(C) 1
(D) -1

Answer Key

Q1	(D)
Q2	(A)
Q3	(B)
Q4	(C)
Q5	(B)
Q6	(A)
Q7	(A)
Q8	(A)
Q9	(D)
Q10	(D)
Q11	(A)
Q12	(D)
Q13	(D)
Q14	(A)
Q15	(B)
Q16	(D)
Q17	(C)
Q18	(C)
Q19	(A)
Q20	(B)
Q21	(B)